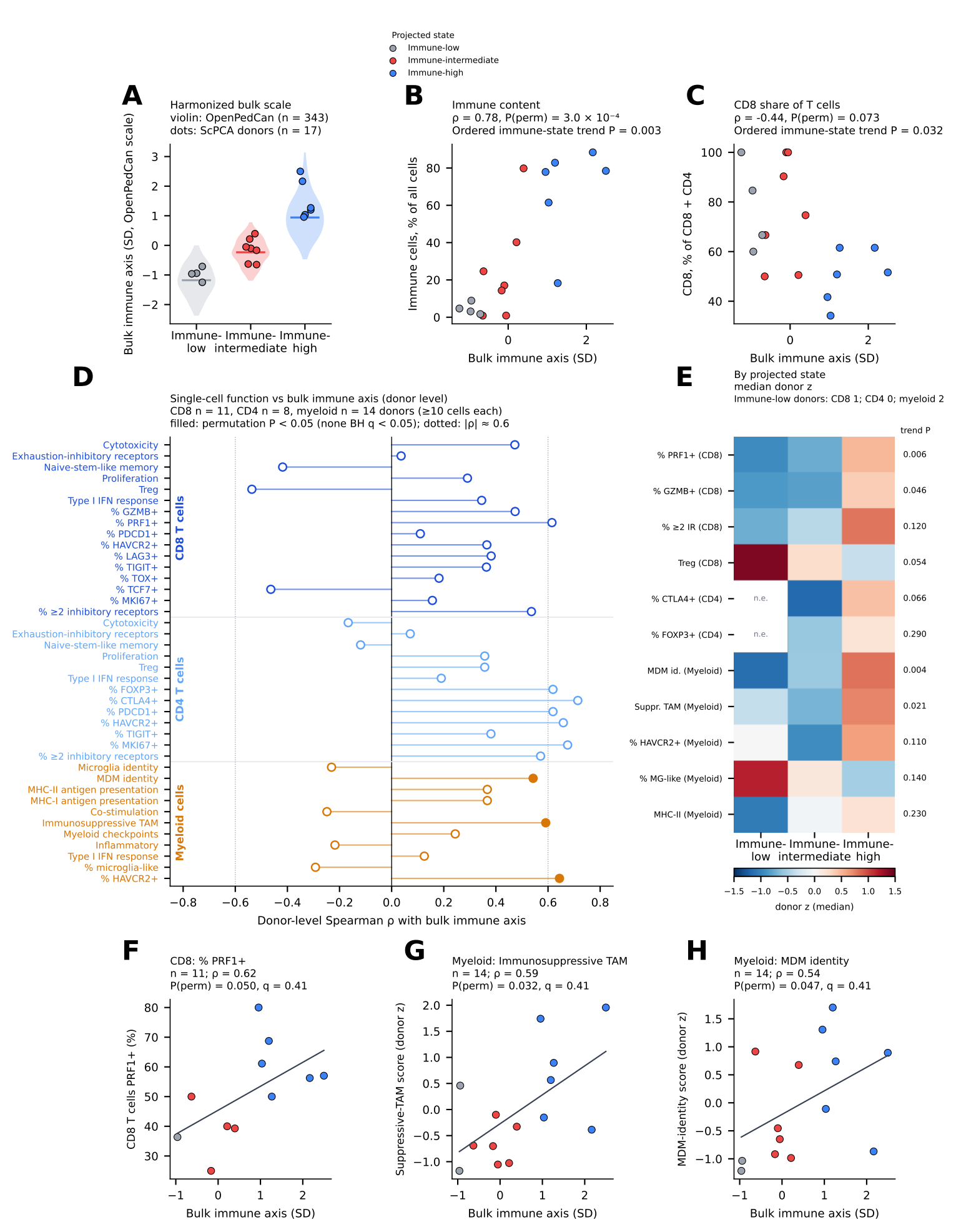

Projected state
Immune-low
Immune-intermediate
Immune-high
A
Harmonized bulk scale
violin: OpenPedCan (n = 343)
dots: ScPCA donors (n = 17)
Bulk immune axis (SD, OpenPedCan scale)
Immune-low
Immune-intermediate
Immune-high
B
Immune content
ρ = 0.78, P(perm) = 3.0 × 10⁻⁴
Ordered immune-state trend P = 0.003
Immune cells, % of all cells
Bulk immune axis (SD)
C
CD8 share of T cells
ρ = -0.44, P(perm) = 0.073
Ordered immune-state trend P = 0.032
CD8, % of CD8 + CD4
Bulk immune axis (SD)
D
Single-cell function vs bulk immune axis (donor level)
CD8 n = 11, CD4 n = 8, myeloid n = 14 donors (≥10 cells each)
filled: permutation P < 0.05 (none BH q < 0.05); dotted: |ρ| ≈ 0.6
CD8 T cells
Cytotoxicity
Exhaustion-inhibitory receptors
Naive-stem-like memory
Proliferation
Treg
Type I IFN response
% GZMB+
% PRF1+
% PDCD1+
% HAVCR2+
% LAG3+
% TIGIT+
% TOX+
% TCF7+
% MKI67+
% ≥2 inhibitory receptors
CD4 T cells
Cytotoxicity
Exhaustion-inhibitory receptors
Naive-stem-like memory
Proliferation
Treg
Type I IFN response
% FOXP3+
% CTLA4+
% PDCD1+
% HAVCR2+
% TIGIT+
% MKI67+
% ≥2 inhibitory receptors
Myeloid cells
Microglia identity
MDM identity
MHC-II antigen presentation
MHC-I antigen presentation
Co-stimulation
Immunosuppressive TAM
Myeloid checkpoints
Inflammatory
Type I IFN response
% microglia-like
% HAVCR2+
Donor-level Spearman ρ with bulk immune axis
E
By projected state
median donor z
Immune-low donors: CD8 1; CD4 0; myeloid 2
trend P
% PRF1+ (CD8)
% GZMB+ (CD8)
% ≥2 IR (CD8)
Treg (CD8)
% CTLA4+ (CD4)
% FOXP3+ (CD4)
MDM id. (Myeloid)
Suppr. TAM (Myeloid)
% HAVCR2+ (Myeloid)
% MG-like (Myeloid)
MHC-II (Myeloid)
n.e.
0.006
0.046
0.120
0.054
0.066
0.290
0.004
0.021
0.110
0.140
0.230
Immune-low
Immune-intermediate
Immune-high
donor z (median)
F
CD8: % PRF1+
n = 11; ρ = 0.62
P(perm) = 0.050, q = 0.41
CD8 T cells PRF1+ (%)
Bulk immune axis (SD)
G
Myeloid: Immunosuppressive TAM
n = 14; ρ = 0.59
P(perm) = 0.032, q = 0.41
Suppressive-TAM score (donor z)
Bulk immune axis (SD)
H
Myeloid: MDM identity
n = 14; ρ = 0.54
P(perm) = 0.047, q = 0.41
MDM-identity score (donor z)
Bulk immune axis (SD)